Dataset Proposal: next-day smog (PM2.5) forecast for Pakistan

HackInsight 2026 · Data-to-Decision · Track: **Air quality** · Team XYZ (Laksh Kumar, Syed Mustehsan Akhtar Kazmi, Noor Fatima, Amna Iqbal) · 21 October 2026
Code: github.com/lazy-laksh/hackinsight-smog-forecast (tag v1.0-final) · Dashboard: huggingface.co/spaces/lazypanda21/smog-forecast-pakistan

> **In one sentence:** we train on four free, credible and independent public sources (ground PM2.5 sensors in the four cities, NASA satellite fire detections, ECMWF-based weather, and the Copernicus CAMS air-quality model) to forecast tomorrow's daily PM2.5 for Lahore, Islamabad/Rawalpindi, Peshawar and Karachi; every raw file is fingerprinted, and the result is checked against Punjab EPA's official stations.

4	**202**	**~905,000**	**~285,000**	**2,500+**
independent sources	PM2.5 sensors in 4 cities	hourly sensor readings	satellite fire detections	raw files with SHA-256

1. Problem, user and what the data must support

Problem: smog decisions in Punjab and Khyber Pakhtunkhwa (school timings, outdoor activity, hospital readiness, enforcement against burning) must be made the evening before, but officials usually have only today's reading. **User:** a district environment officer. **Decision:** "Do I issue a smog advisory for tomorrow?" at 20:00 PKT.

So the data must give (a) a measured PM2.5 target for each city and day, (b) the drivers of smog that are known by the evening: pollution already in the air, crop-burning fires, weather that traps or clears smog, and (c) enough history to learn a full smog season. Smog in these cities is driven by local emissions, crop-residue fires and stagnant weather; our four sources cover the effect and each main cause.

2. The sources used for training

Source	Publisher and access	What we use	Coverage and volume	Role	Licence
OpenAQ PM2.5 sensor network	OpenAQ platform, aggregating each sensor's owner (see section 3). API v3, api.openaq.org/v3, free key	Hourly PM2.5 from every sensor within 25 km of each city centre	202 sensors (Lahore 87, Karachi 53, Islamabad/Rawalpindi 39, Peshawar 23); June 2025 to now; ~905,000 sensor-hours	**The target** and "smog already in the air" inputs	Per sensor, mostly CC BY 4.0; recorded per file in manifest.csv
NASA FIRMS active fires	NASA ESDIS / LANCE. firms.modaps.eosdis.nasa.gov/api, free MAP_KEY. DOI 10.5067/FIRMS/VIIRS/VJ114IMGT_NRT.002	VIIRS NOAA-20 375 m fire detections in a box covering Pakistani and Indian Punjab, Sindh and KP	~285,000 detections, June 2025 to now, near-real-time (~3 h)	Crop-burning inputs: fires within 100 and 300 km	NASA open data policy; acknowledgement in README

Open-Meteo weather	Open-Meteo, from ECMWF ERA5 / IFS. open-meteo.com, no key	Hourly wind, rain, humidity, temperature, pressure, cloud and boundary-layer (mixing) height; archive plus forecasts	~48,800 city-hours plus hourly forecast snapshots	Weather inputs, including tomorrow's forecast	CC BY 4.0 ("Weather data by Open-Meteo.com")
Copernicus CAMS air-quality model	Copernicus Atmosphere Monitoring Service (EU), served by Open-Meteo. air-quality API	Modelled PM2.5, dust and aerosol optical depth	~48,800 city-hours	**Input only**, never the target or used for evaluation	CC BY 4.0, attribution to CAMS and Open-Meteo

Why each source is credible

- **OpenAQ** is a non-profit, open air-quality data platform used by researchers and governments; it records each sensor's owner, instrument and licence. In our four cities most sensors belong to Pakistani organisations (section 3).
- **NASA FIRMS** is NASA's official fire product, has DOIs and is widely used in peer-reviewed research on crop-residue burning and smog in South Asia. It is a satellite observation, fully independent of the ground sensors.
- **Open-Meteo** serves data from **ECMWF**, the European Centre for Medium-Range Weather Forecasts, the world's leading weather-forecasting centre.
- **CAMS** is the European Union's Copernicus atmospheric service; a physics model of how pollution forms and moves. We use it as a hint, so our target stays real measurements.

3. Who owns the sensors behind our target

OpenAQ only aggregates; the measurements come from each sensor's owner. Of our 202 sensors, **155 belong to Pakistani organisations**:

Owner (as recorded by OpenAQ)	Instrument provider	Sensors (Lahore / Karachi / Isb-Rwp / Peshawar)
Pakistan Air Quality Initiative (PAQI)	Hawanama	**88** (49 / 14 / 16 / 9)
The Urban Unit (Government of Punjab company)	AirGradient	**67** (21 / 18 / 15 / 13)
OpenAQ (platform-managed)	Hawanama	17 (8 / 9 / 0 / 0)
AirGradient	AirGradient	16 (6 / 7 / 3 / 0)
US diplomatic-post reference monitors (StateAir / AirNow)	StateAir, AirNow	5 (1 / 1 / 2 / 1)
Clarity	Clarity	2 (1 / 1 / 0 / 0)
Individuals or unlabelled owners	AirGradient	7

Counts from the sensors table published on our dashboard ("About the data" tab), 21 Oct 2026.

4. How the data is used in model training

Item	What we did
Target	Tomorrow's mean PM2.5 (µg/m³) for each city on the Pakistan calendar day: the mean of hourly city medians of usable sensors, needing at least 18 of 24 hours. Secondary target: whether that mean is ≥ 55.5 µg/m³ (US EPA "Unhealthy").
Inputs (known at 20:00 PKT)	Today's PM2.5 so far and this evening; yesterday and earlier days; today's weather and tomorrow's weather forecast; fires within 100 and 300 km (today and last 3 days); CAMS PM2.5 for today and tomorrow; season, day of week, city. A test proves no input uses tomorrow's real readings.

Training period	Target days 2 June 2025 to 30 September 2026: **1,922 city-days**, including the full 2025-26 smog season.
Model selection	Rolling-origin validation (train on the past, predict the next month, repeat): **1,458 city-days**. Settings and the 80% range calibration were chosen here only.
Locked test	Every target day from **1 October 2026** (80 city-days, 1–20 Oct), never used for tuning.
Baseline and success (declared 19 Oct, before results)	Persistence ("tomorrow = today"), MAE, at least 10% better overall and for Lahore. Result: **17.7% better overall** (MAE 8.94 vs 10.86); Lahore 8.7% (target not met, reported openly).

5. Data quality and provenance

- **Provenance:** all data is downloaded by our own code. Every raw file is listed in `data/manifest.csv` with source, exact query (keys hidden), fetch time (UTC), SHA-256 fingerprint, size, licence and publisher; the fingerprints are re-checked every hour.
- **Layers:** raw (never edited) → cleaned (typed tables, UTC) → gold (city-hour and city-day tables, each row with an ID that the Insight Agent cites).
- **Data Quality Agent:** flags zero, stuck (6+ identical hours), spiking and impossible readings; fills only 1–2 hour gaps (marked as filled); logs every decision in `data/logs/quality_log.csv`; two gates stop publishing if the data is malformed, stale or incomplete. Example run: 1,232 zero hours, 292 stuck hours and 28 spikes flagged; 27,919 short gaps filled.
- **Known gaps:** about a dozen OpenAQ sensor-months return server errors; they are logged in `openaq_known_gaps.csv`, never invented. VIIRS SNPP was missing for May 2026, so only NOAA-20 is used for a consistent fire count.
- **Reproducible:** one command, `make data` (or `python -m smog data`), rebuilds everything; tested from an empty folder on a second laptop (all 8 stages passed, about 65 minutes, mostly API speed limits).

6. Pakistani official sources used for validation and reference

Source	Publisher	How we use it
Punjab AQI portal, government reference stations	Environment Protection Agency, Punjab	**Local validation only** (no data licence stated, so not used for training): our Lahore value was 10.2 µg/m³ lower than the EPA station mean on average over 4 hours, always inside the stations' range
National Environmental Quality Standards for Ambient Air, S.R.O. 1062(I)/2010	Government of Pakistan	Pakistan's legal PM2.5 limit (35 µg/m³ per 24 h), cited by the Insight Agent
Punjab smog mitigation results 2025	Environment Protection & Climate Change Dept., Punjab	Problem evidence (Lahore peaks 539 and 237 µg/m³)
Punjab Smog Action Plan 2025-26	Planning & Development Board, Punjab	Shows the official need for AI-based early warning
Pre-Smog Action Plan for schools 2026	Dawn, reporting a Punjab School Education Dept. notification	What schools must do; cited by the agent
WHO Air Quality Guidelines 2021; US EPA AQI PM2.5 breakpoints 2024	WHO; US EPA	Health thresholds and the "Unhealthy" line (55.5 µg/m³)

7. Considered and not used

- **US embassy / consulate monitors (AirNow):** reference-grade, but the public historical download was removed in March 2025, so they cannot support a live hourly pipeline; the few that appear on OpenAQ are included there.
- **Punjab EPA stations for training:** no download or stated data licence; used for validation only.

- **IQAir / AQICN:** terms do not allow free reuse in a product; they re-publish other sources.
- **Sentinel-5P satellite (NO2, aerosol):** daily, coarse and often hidden by cloud and haze; too heavy for 72 hours.

8. Rules, limits and ethics

- No personal or identifiable data; no synthetic or AI-generated data for training or evaluation.
- Licences respected and recorded per file; attributions shown in the README and on the dashboard. Open-Meteo's free API is non-commercial, so a paid plan is needed before charging customers.
- Limits: most target sensors are low-cost optical sensors without humidity correction (about 11% below official stations in our spot check); more sensors in central areas; satellites miss fires under cloud; weather grid cells are kilometres wide. These are stated in the model card and datasheet (`docs/DATASHEET.md`).

Full details: DATASET_DECLARATION.md (declared 19 Oct 2026), docs/DATASHEET.md, reports/MODEL_CARD.md, data/manifest.csv in the repository.